its wave function obeys the Schrödinger equation, undulating and propagating outward like a wave. But when the electron hits the phosphorescent screen, it hits in one location, lighting up a particular spot on the screen, like a particle. So sometimes the electron behaves like a wave, and sometimes it behaves like a particle, but never both. According to Bohr, there cannot be a more complete description of an electron, or of anything—merely incomplete and incompatible analogies that never overlap. This, Bohr said, was the heart of complementarity, and it was inevitable and unavoidable. The new quantum theory had shown it was impossible to give a single consistent account of an electron that would work at all times.

Bohr pointed to the Heisenberg uncertainty principle as further justification for the inevitability of complementarity. Using Heisenberg's gamma-ray flashlight as an example, he pointed out that there was no way to avoid altering the momentum of an electron when observing its position, and vice versa. Bohr then echoed Mach, as Heisenberg had, and claimed the impossibility of measuring both properties of an electron simultaneously meant that it could not have both properties simultaneously. Position and momentum, like particles and waves, were complementary—never used at once but both needed for the complete description of a situation.

But Bohr was wrong. There was nothing inevitable or necessary about complementarity—other interpretations of quantum physics are possible. Indeed, the claim of inevitability is an awfully strong and strange claim to make about any interpretive issue in science, precisely because it is always possible to reinterpret any theory. Yet Bohr was convinced that complementarity was the deepest insight into nature found within the quantum theory.

Stranger still is Bohr's use of the gamma-ray flashlight to bolster his claims. It's certainly true that the thought experiment illustrates a world in which there are limits on our knowledge, but it's also a world where particles have well-defined positions and momenta at all times. Hitting an electron with a gamma ray can't alter the electron's momentum unless it has a momentum in the first place. We don't know what that momentum is—but that's certainly not the same thing as saying it doesn't exist.

As is always the case with Bohr, it's hard to be sure what he was actually trying to say, because his writing is so convoluted and obscure. But